

Exact degree-ten invariants of a self-dual five-form in ten dimensions

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(Dated: August 3, 2026)

The Lorentz-invariant scalar functions of a self-dual five-form in ten dimensions control the non-linear self-interactions the field can have. We determine the degree-ten sector of this structure exactly over $\mathbb{Q}$, rather than modulo a prime. The full invariant space has dimension 14; the subspace activated by the stress-tensor flow has dimension 11; the quotient therefore has dimension 3, so three independent degree-ten directions lie outside the reach of the flow. The separation rests on a short theorem: the free stress tensor of a self-dual five-form is traceless for every improvement coefficient, because $F \wedge F$ vanishes identically for a form of odd degree, so $\text{Tr } \tau$ begins at field degree four rather than two. Forcing the naive degree-two start collapses the quotient to zero. We certify a tensor–spinor bridge of rank 126 with an exact left inverse, certify generic functional rank at least 81 by an explicit 81×81 integer minor, and give the one-dimensional overlap between the published degree-ten span and the product sector as an exact integer identity. All computations are exact; none uses floating point.

**DRAFT FOR MENTOR REVIEW — NOT FOR
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Introduction.—A chiral p -form has a field strength constrained to equal its own Hodge dual. The constraint is forced on us in string theory and sits badly with the usual machinery: a self-dual field strength admits no manifestly Lorentz-invariant action of the naive kind, a difficulty identified by Marcus and Schwarz [1] and addressed since by a sequence of constructions [2–7]. The obstruction is visible in one line, and we recover it below as (8): the candidate kinetic term $\langle F, F \rangle$ vanishes identically on the constrained surface.

Asking not for a chiral theory but for the space of non-linear self-interactions one can have turns the problem into a question about scalar invariants. In four dimensions the answer is classical and the duality-invariant extensions of Maxwell theory are completely characterised [8, 9]; in six dimensions the self-interacting chiral two-form and its $T\bar{T}$ -like stress-tensor flow are well developed [10–12]. Ten dimensions is harder for combinatorial rather than conceptual reasons: the self-dual five-form lives in a 126-dimensional module and the contraction patterns available at degree ten are too numerous for reliable hand enumeration. Explicit degree-ten structures and the relevant flow appear in [13, 14]. What has been missing is the exact dimension of the degree-ten invariant space, the exact dimension of the part the flow reaches, and the relation between them. We supply all three.

The five-form is not an arbitrary choice of example. It is the Ramond–Ramond field strength of Type IIB supergravity, whose covariant field equations are due to Schwarz [15] and for which manifestly Lorentz-invariant actions were later built with precisely the chiral-boson machinery cited above [16]; higher-derivative cor-

rections involving it have been studied directly [17], within the eight-derivative programme [18, 19], and non-perturbatively through D-instantons [20]. We make no Type IIB claim here, and say so again below; the point is only that the invariant structure of this particular module is not an idle question.

Two features of the method are worth stating at the outset, because they are what the results rest on. First, everything is computed in exact arithmetic: integer or rational, or modular over $\mathbb{F}_p$ with the passage to characteristic zero made explicit each time. No quantity in this paper is a floating-point number and no rank is obtained by thresholding a singular value. Second, the enumeration is carried out twice, in two languages that share no evaluation code, so that agreement between them is informative rather than tautological. The general enumerate–evaluate–relate workflow is prior work [21]; what we contribute is its exact ten-dimensional realisation, the certified bridge, and the characteristic-zero certificates.

Setup.—Work on $\mathbb{R}^{1,9}$ with $\eta = \text{diag}(-1, +1, \dots, +1)$ and $\epsilon_{01\dots 9} = +1$. On five-forms the Hodge star acts by $(\star F)_{\mu_1\dots\mu_5} = \frac{1}{5!} \epsilon_{\mu_1\dots\mu_5}{}^{\nu_1\dots\nu_5} F_{\nu_1\dots\nu_5}$. For a p -form in $d = 2p$ dimensions with t timelike directions,

$$\star^2 = (-1)^{p(d-p)+t} = (-1)^{p^2+t}, \quad (1)$$

so at $p = 5$, $d = 10$ one has $\star^2 = -(-1)^t$: eigenvalues ± 1 in Lorentzian signature and $\pm i$, with no real eigenspace decomposition, in Euclidean. Self-duality of a five-form is thus a Lorentzian phenomenon. The projectors $P_{\pm} = \frac{1}{2}(\mathbf{1} \pm \star)$ satisfy $P_{\pm}^2 = P_{\pm}$ and $\text{tr } P_{\pm} = 126$, and Λ^5 splits as $\Lambda_+^5 \oplus \Lambda_-^5$ with $\dim \Lambda_{\pm}^5 = 126$ inside $\dim \Lambda^5 = 252$. Write $F \in \Lambda_+^5$. “Degree” always means degree in F ; only even degrees occur, and A_d denotes the invariants of degree d .

The invariants themselves are built by full contraction. A degree- d candidate is specified by a contraction graph: d copies of F carrying $5d$ indices, paired off by η and, where the pattern demands it, by one ϵ . Each graph evaluates to a polynomial with *integer* coefficients in the

126 coordinates, fixed before any prime is chosen, since η and ϵ have integer entries; the prime enters only when that fixed polynomial is evaluated. This is what makes the spanning-set argument available: a space spanned by exactly k explicit elements has $\dim_{\mathbb{Q}} \leq k$ for free, so a modular rank of k pins the rational dimension with no prime excluded. It applies to A_{10} and to the product sector, and it is precisely what is *not* available for D_{10} , whose elements the closure produces rather than receives — which is why that space is treated by exact lifting instead.

The Lorentz group acts on Λ_+^5 with $\dim \mathfrak{so}(1, 9) = 45$. A generic point has a stabiliser, and the number of functionally independent invariants at a generic point is 126 minus the generic orbit dimension. That analytic count is 81 and is not ours [14]; what we contribute is an exact computational lower bound matching it.

The tensor-spinor bridge.—Two independent constructions are used throughout, one from contraction graphs and one from chiral bispinors, and their agreement is the paper’s main internal control. They are related by

$$\Phi(F)_{ab} = \frac{1}{5!} F_{\mu_1 \dots \mu_5} (C \Gamma^{\mu_1 \dots \mu_5})_{ab}, \quad (2)$$

with $a, b = 1, \dots, 16$. The target is not all of $\text{Sym}^2 S_+$: with $\dim S_+ = 16$ we have $\dim \text{Sym}^2 S_+ = 136$, the gamma trace $(\gamma\text{-tr } \psi)_{\mu} = \eta_{\mu\nu} (\Gamma^{\nu})^{ab} \psi_{ab}$ has rank 10, and

$$\dim \ker(\gamma\text{-tr}) = 136 - 10 = 126 = \dim \Lambda_+^5. \quad (3)$$

Proposition 1 (certified). *Over $\mathbb{F}_p$ at each prime used, $\text{im } \Phi = \ker(\gamma\text{-tr})$, $\text{rank } \Phi = 126$, $\ker \Phi \cap \Lambda_+^5 = 0$, and $\Phi|_{\Lambda_+^5} \equiv 0$. There is an exact left inverse with $\Phi^- \Phi = \text{id}$, while $\Phi \Phi^-$ is the projector onto $\ker(\gamma\text{-tr})$ and not the identity. Equivariance holds with the intertwining character solved for, not assumed, and found to be unity.*

Two points are easy to get wrong. Φ has full column rank but is not onto, so it admits a left inverse and no two-sided one; both identities are verified as matrix statements, since one that holds on sampled vectors can still fail on a subspace. And the vanishing on Λ_+^5 is exact, not approximate: on S_+ the anti-self-dual part of $\Gamma^{(5)}$ annihilates the module. That blindness makes an orientation error silent. The null frame over $\mathbb{F}_p$ solves $L^T \eta L = \eta_{\text{null}}$, which fixes L only up to square-root branches; two solutions differing in $\det L$ induce $\star \mapsto \pm \star$ and hence rank 126 versus rank 0, with no error raised either way. We pin the branch by construction and verify $\star F = +F$ on the image — a property of the output rather than of the criterion that selected it. The defect was found because three cells of the rank matrix failed at a single prime, and was diagnosed as a branch rather than an exceptional prime by flipping it deliberately at both a failing and a working prime. No published number changed.

One further point deserves care, because an earlier version of this project recorded its opposite. The spinor computation is natural in a null frame adapted to split

signature $(5, 5)$, while the physical problem is Lorentzian. These are *not* related by a real orthogonal transformation — a real congruence preserves signature, so no $\Lambda \in O(1, 9; \mathbb{R})$ carries $\eta_{(1,9)}$ to $\eta_{(5,5)}$. They are different real forms of one complex algebra,

$$\mathfrak{so}(1, 9)_{\mathbb{C}} \cong \mathfrak{so}(10, \mathbb{C}) \cong \mathfrak{so}(5, 5)_{\mathbb{C}}, \quad (4)$$

and every dimension, rank and intersection quoted here is the dimension of a $\mathbb{C}$ -linear object and so is constant along (4). What (4) does *not* transport is which eigenspace of $\star$ survives: $\star$ is built from η , and by (1) the reality of its eigenspaces is exactly a signature-dependent statement.

The trace theorem.—Everything at degree ten turns on the field degree at which $\text{Tr } \tau$ begins.

Theorem 2. *For a self-dual five-form in ten dimensions the free stress tensor is traceless, for every improvement coefficient. Hence $\text{Tr } \tau$ first contributes at field degree four.*

Proof. Take the general improved form

$$T_{\mu\nu} = \frac{1}{(p-1)!} F_{\mu\alpha_2 \dots \alpha_p} F_{\nu}{}^{\alpha_2 \dots \alpha_p} - c \eta_{\mu\nu} \langle F, F \rangle, \quad (5)$$

with $\langle F, G \rangle = F_{\mu_1 \dots \mu_p} G^{\mu_1 \dots \mu_p}$ and c free. Contracting with $\eta^{\mu\nu}$,

$$T^{\mu}{}_{\mu} = (1 - cd) \langle F, F \rangle, \quad (6)$$

a multiple of the single scalar $\langle F, F \rangle$ whatever c is: no other quadratic Lorentz invariant exists, the full contraction being the only way to pair two p -forms into a scalar. Now specialise to $p = 5$, $d = 10$, so F has odd form degree. Exchanging two factors of a q -form costs $(-1)^{q^2} = (-1)^q$, which is -1 at $q = 5$, so

$$F \wedge F = -F \wedge F \implies F \wedge F \equiv 0 \quad (7)$$

for every five-form. Since $\langle F, \star F \rangle \text{vol} \propto F \wedge F$, the pairing $\langle F, \star F \rangle$ vanishes, and on either eigenspace $\star F = \pm F$ gives

$$\langle F, F \rangle = \pm \langle F, \star F \rangle = 0. \quad (8)$$

Substituting into (6) gives $T^{\mu}{}_{\mu} = 0$ for all c . $\square$

The two steps are independent, and that matters for how the statement is tested. Equation (7) is about form parity and holds for any five-form; self-duality enters only at (8). A suite checking self-dual samples alone would confirm the conclusion without exercising the mechanism, so we check both eigenspaces — the argument predicts vanishing on each — and add a control confirming $\langle F, F \rangle \neq 0$ on an *unprojected* five-form, without which the other checks pass vacuously. The sign in (1) is likewise asserted by test wherever the decomposition is used: a Euclidean projector, where $\star^2 = -1$, was once mistaken

for an anti-self-dual one here and its non-vanishing briefly read as signal.

Activation and the exact closure.—The flow generates new invariants by repeatedly applying scalars built from traces of powers of τ . Through degree ten the generators are $\text{Tr } \tau^k$ for $k \leq 5$ together with the products $(\text{Tr } \tau)^2$, $\text{Tr } \tau \text{ Tr } \tau^2$, $\text{Tr } \tau \text{ Tr } \tau^3$, $(\text{Tr } \tau^2)^2$ and $\text{Tr } \tau^2 \text{ Tr } \tau^3$; the list was obtained by instrumenting the closure, so that a generator consumed but undocumented would appear. Writing $\deg_F$ for leading field degree,

$$\deg_F \text{Tr } \tau^k = \begin{cases} 4, & k = 1, \\ 2k, & k \geq 2, \end{cases} \quad (9)$$

with degrees adding on products. The exceptional case is Theorem 2: the naive count gives 2, and the vanishing of the free trace pushes it to 4.

A target is *generated* when the flow produces it formally and *activated* at degree d when (9) lets it contribute there. Concretely, the reachable space at degree d is the least fixed point of

$$V \mapsto V + \text{span}\{g : g \text{ generated from } V, \deg_F g \leq d\}, \quad (10)$$

started from the degree-four seed and iterated until a sweep raises no rank. Over $\mathbb{Q}$ the reachable dimensions by degree are 1, 2, 7 and 11 at degrees four, six, eight and ten.

The distinction between generated and activated is the whole result. Writing $\mathcal{T}$ for the generated targets and $\mathcal{A}$ for activation,

$$\text{span } \mathcal{T} = A_{10} \quad \text{but} \quad D_{10} = \text{span} \bigcup_{k \geq 0} \mathcal{A}^k(\mathcal{T}_0) \subsetneq A_{10}. \quad (11)$$

The left equality is why conflating them is tempting; doing so returns quotient dimension zero, which is the answer this project held before catching the error.

Running the closure modulo p would expose the fixed point to exceptional primes precisely where no spanning-set argument is available, so it is run over $\mathbb{Q}$. Of its target rows 49 were already integral; the remaining 9 were lifted by the Chinese remainder theorem across 5 primes with rational reconstruction [22] and validated at the held-out prime 32771, with no mismatches. This is not fastidiousness. Figure 1 plots the size of the largest coefficient the closure actually produces at each degree. Through degree eight the coordinates are small integers and nothing is at risk. At degree ten one basis vector of D_{10} carries a numerator of twenty-nine bits over a five-digit denominator — an order of magnitude past what a single prime near 2^{15} can represent. Read modulo one such prime that coefficient is indistinguishable from a small residue, and the fixed point would have been reported at the wrong dimension with nothing in the output to indicate it. Iteration terminated after 3 sweeps.

Theorem 3. *Over the rationals,*

$$\dim_{\mathbb{Q}} A_{10} = 14, \quad \dim_{\mathbb{Q}} D_{10} = 11, \quad \dim_{\mathbb{Q}} Q_{10} = 3. \quad (12)$$

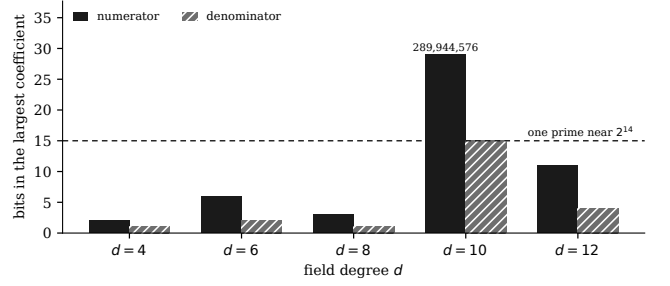


FIG. 1. Largest numerator and denominator in a basis vector of D_{10} , by field degree, in bits. The dashed line is the range of a single prime near 2^{15} . Degree ten is the first place the exact coefficients leave it, which is why that row is lifted rather than read off one modular solve.

The lower bound is certified by an explicit 11×11 integer minor with non-vanishing determinant, computed by Bareiss elimination [23] and confirmed by an independent exact rational routine. No separate upper bound is needed: over $\mathbb{Q}$ the modular admission test that made the predecessor a bound disappears, and at the fixed point no further generated direction raises the rank, so the reached span is D_{10} and $\dim_{\mathbb{Q}} D_{10} = 11$ is an equality. Three exact annihilating covectors give the quotient map directly. Forcing $\text{Tr } \tau$ to begin at degree two instead activates 24 further targets, raises the reachable dimension to 14, and returns $Q_{10} = 0$; that counterfactual is retained as a regression fixture.

How many directions must be added to close the gap is a basis-dependent question only in appearance.

Proposition 4. *If $S \subset A_{10}$ satisfies $D_{10} + \text{span}(S) = A_{10}$, then $|S| \geq \dim Q_{10} = 3$.*

Proof. Apply the quotient map $\pi : A_{10} \rightarrow Q_{10}$. From $D_{10} + \text{span}(S) = A_{10}$ we get $\text{span}(\pi(S)) = Q_{10}$, and a span of $|S|$ vectors has dimension at most $|S|$. $\square$

This is elementary and we claim no novelty for it; we state it because it converts a basis-dependent assertion into a basis-free one. No closing set of any kind, in any basis, has fewer than 3 elements. Removal minimality of the particular closing set we exhibit — that no element can be dropped — is checked only in a fixed graph basis and under the permutation subgroup, and we do not claim it under arbitrary GL .

Degree-resolved agreement.—The two formulations are compared on 78 common samples at prime 32749. Ranks agree at degrees four, six, eight and ten [Fig. 2(c)], which is meaningful corroboration precisely because a shared omission would have to occur twice in two different languages. At degree eight the full spinor family attains 7; deleting its tensor words drops it to 6, so within this decomposition the tensor words are indispensable. We state that in the qualified form the evidence supports: it is a statement about the families constructed here, not a uniqueness theorem.

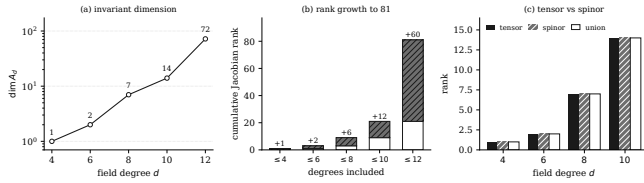


FIG. 2. (a) invariant dimension by field degree; (b) cumulative Jacobian rank, with the per-degree increment annotated, saturating at 81; (c) tensor, spinor and union ranks, equal at every degree shown.

Generic functional rank.—Let $I_1, \dots, I_N$ be the candidates, $N = 83$, in the $n = 126$ coordinates F^A . The generic functional rank is the rank at a generic point of $J_i^A = \partial I_i / \partial F^A$, an 83×126 matrix. Rows are differentiated analytically, by amputating one F from the contraction pattern, so there is no step size; each is checked against Euler’s relation $\sum_A F^A \partial_A I_i = d_i I_i$, which ties a differentiated row to an undifferentiated evaluation and so detects exactly the errors differentiation introduces. All 83 of 83 candidates were evaluated, with 0 errors and 0 zero rows.

The exact modular rank is 81, and an explicit 81×81 minor has non-zero determinant. A non-vanishing minor over $\mathbb{F}_p$ lifts to one over $\mathbb{Q}$, since the determinant is an integer non-zero mod p ; hence

$$\text{generic functional rank} \geq 81 \quad (13)$$

with no prime excluded. Repeating across 5 primes and 3 seeds gives 15 cells, every one at rank 81 with the same 81 pivot rows and 81 pivot columns and 0 unstable. Two matrices can share a rank and disagree about which rows realise it, so the stability of the pivot *sets* is the stronger statement and is the one we record. We are nonetheless deliberately restrained: cells from one run agreeing with each other is close to guaranteed whether or not the run is correct, so the matrix corroborates and the minor is what carries the bound.

The rank is also not concentrated at one degree. Sorting the 81 pivot rows of the certified minor by the degree of the candidate that supplies them gives 1, 2, 6, 12 and 60 at degrees four through twelve, and sorting them by construction gives 68 port graphs against 9 tensor words and 4 spinor candidates. The tensor words carry pivots of their own, which is the same conclusion the degree-eight ablation reaches by a different route.

The matching *upper* bound is analytic and is not ours [14]; we claim the lower bound and the certificate. We do not claim that all 83 candidates are algebraically independent.

The published span and the product sector.—Let B_{10} be the span of the published degree-ten structures and P_{10} the product sector. Over $\mathbb{Q}$, $\dim B_{10} = 12$, $\dim P_{10} = 2$, $\dim(B_{10} + P_{10}) = 13$ and

$$\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1, \quad (14)$$

TABLE I. Degree-resolved data. Ranks are exact modulo 32749; dimensions of A_{10} and D_{10} are exact over $\mathbb{Q}$.

d	tensor	spinor	union	$\dim_{\mathbb{Q}} D_{10}$ at d
4	1	1	1	1
6	2	2	2	2
8	7	7	7	7
10	14	14	14	11

generated by the exact integer identity

$$\begin{aligned} & -55 I^{(1)} - 252 I^{(2)} + 4200 I^{(5)} + 5040 I^{(6)} \\ & = -1040 I_1^{(4)} I_1^{(6)}, \end{aligned} \quad (15)$$

verified at the fresh prime 32869 on 6 samples that took no part in the reconstruction. The coefficients are normalisation-dependent, unlike every dimension quoted above; what is normalisation-independent is that the intersection is one-dimensional and supported on the four published structures listed. We read this as a structural observation about the basis rather than a defect in it.

What the quotient means, and what it does not.—It is worth separating the arithmetic from its reading. That $\dim_{\mathbb{Q}} Q_{10} = 3$ is certified; what that number licenses one to say is a separate question, and we are deliberately sparing. The flow of [14] deforms a chiral theory along a direction built from its own stress tensor, in the manner of [24–27] in two dimensions. A non-zero quotient says that at degree ten the deformations this flow generates do not exhaust the invariants a chiral five-form admits: there are three independent directions in which the theory can be deformed, consistently with Lorentz invariance and self-duality, that the flow does not produce from its seed. Whether those directions correspond to sensible interacting theories is not settled here. Perry and Schwarz [28] and Bekaert and Henneaux [29] showed that self-interactions of a chiral form are constrained beyond Lorentz invariance — the latter deriving the condition from the Dirac–Schwinger requirement on the energy–momentum commutator — and we impose no such condition. The three directions are invariant-theoretic, and calling them physical would be a further claim requiring further work.

Scope.—Every dimension here is exact for the span of the enumerated family. Reading $\dim A_{10} = 14$ as the dimension of *all* degree-ten Lorentz invariants further requires the enumeration to miss nothing, which we do not prove; the two independent constructions agreeing at four degrees is corroboration, not a theorem, and a Hilbert-series or character count would settle it. We do not prove full degree-twelve tensor–spinor equivalence, give a complete invariant-ring presentation, claim canonicity of any basis, or claim minimality under arbitrary GL change of basis. No Type IIB prediction, no statement about causal propagation or hyperbolicity, and no supersymmetry claim is made or implied.

Reproducibility.—All computations are exact and none uses floating point. 254 tensor-side and 86 bridge-side tests pass from a clean clone, and the certificates, figures and tables regenerate from committed scripts; no scientific value is typed into the manuscript source, each being read from a generated macro so that the text and the artifacts cannot drift apart. Cells of the rank matrix are written once and aggregated read-only, with the aggregation independent of order, and a set of gates runs against the compiled source to check specific wording and consistency requirements — that no unscoped completeness claim appears, that a quantity certified as a bound is never described as an equality. Each gate was negative-tested by deliberately violating the condition it detects, since a gate that cannot fail is not a gate.

Two defects were found only by clean-clone reproduction, and we record them rather than quietly fixing them. One was an undocumented Python floor. The other is more instructive: the modular contractor enforces an intermediate-size and a floating-point-operation budget, and both checks sat behind an optional import that the reproduction documents described as not installed for the certified runs. In that configuration neither budget was ever evaluated, and the guard against runaway contractions was inactive in exactly the environment a reproducer was told to use. The dependency is now required and the fallback raises rather than running unguarded; no computed value changed.

We also state a limit on reproducibility rather than leaving it to be inferred. Figures, generated tables and the source archive are byte-identical across clean rebuilds within a pinned toolchain. The compiled PDF is not: the typesetter writes fresh build metadata on every invocation, and three consecutive compilations of identical source produce three distinct digests. The hash recorded for a PDF identifies one build; it does not certify a reproducible one.

Conclusions.—The degree-ten Lorentz-invariant structure of a self-dual five-form in ten dimensions is now known exactly over $\mathbb{Q}$. The full invariant space has dimension 14, the stress-flow-activated subspace has di-

mension 11, and the quotient has dimension 3. The published degree-ten span meets the product sector in exactly one dimension, written out as an explicit integer identity. Generic functional rank is at least 81, certified by an explicit minor rather than by an elimination whose correctness one would have to take on trust.

The separation of D_{10} from A_{10} rests entirely on Theorem 2, and the theorem is short: the free stress tensor of a self-dual five-form is traceless because $F \wedge F$ vanishes identically for a form of odd degree, so $\text{Tr } \tau$ begins at field degree four. That shortness is a virtue here. The argument uses no property of the particular free dynamics, no choice of improvement term and no normalisation, so it survives reformulation of the theory. What it does assume is that the object appearing in the flow is a stress tensor of the shape (5). Whether that holds in the formulation the source intends is a question about conventions rather than about our computation, and we have flagged it rather than assumed the favourable reading, because the counterfactual is not a small correction: it takes the quotient from 3 to zero and changes what the paper says.

Three directions follow. A Hilbert-series or character-theoretic count of the degree-ten invariants of the 126-dimensional module would settle the exhaustiveness question that our two independent enumerations only corroborate, and is the single most valuable check anyone could add. Extending the equivalence to degree twelve requires the bridge argument at a degree where the candidate count grows sharply and where we currently have only partial certified input. And the physical status of the three unreached directions — whether they support consistent interacting chiral theories, or are obstructed by the Dirac–Schwinger condition that constrains chiral self-interactions generally — is open, and is the question we would most like answered.

ACKNOWLEDGMENTS

[Pending author decision.]

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